

The statistics of soaring flight

EconophysicsLab offsite 2026



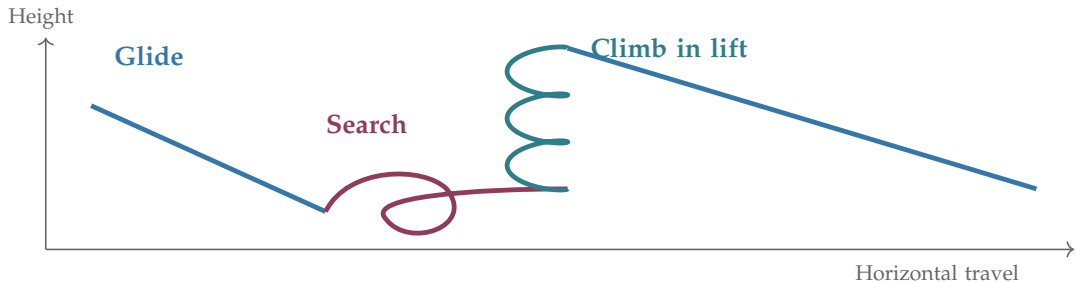
Matteo Di Sante

Transport, routes and
launch environments

Soaring is transport with an energy budget

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Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Schematic

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

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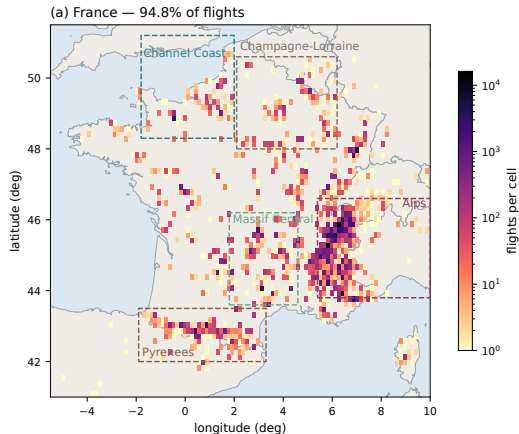
FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

Cleaning and filtering: the full procedure

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1 Altitude

GNSS throughout.
Check presence and range.

2 Fixes

Remove impossible positions;
invalidate bad altitudes.

3 Airborne interval

Trim ground phases;
split witnessed ground stops.

4 Flight selection

Duration ≥ 40 min
Path ≥ 20 km; altitude range
 ≥ 600 m
Integrity and reach checks

5 Coordinates

Local east and north.
Keep GNSS altitude.
One origin per flight.

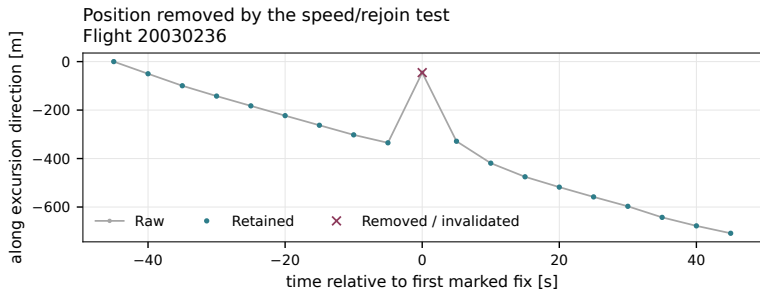
6 Time grid

Fill short gaps; split long gaps.
Flag repairs; smooth
and differentiate.

A local outlier flag alone never deletes a fix; impossible steps drive removal

Retain supported motion, record every repair

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Real example:
remove an impossible excursion when the track rejoins.

Long unsupported intervals
become boundaries.

156,406

84.1% retained

Paragliders

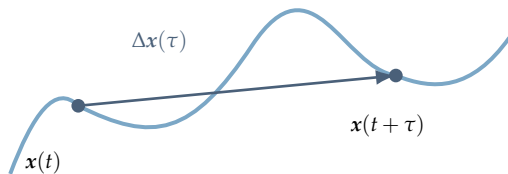
6,094

90.7% retained

Hang gliders

Paraglider grid reconstruction: 0.58%

Hang glider: 0.30%



Displacement is the straight-line change in position, not distance travelled.

Average within each flight, then give each flight equal weight.

$$m_f(\tau) = \frac{1}{n_f} \sum_i |\Delta x_{fi}|^2$$

$$M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$$

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusion} \quad H = 1 \text{ ballistic motion}$$

Keep the same flights at every lag

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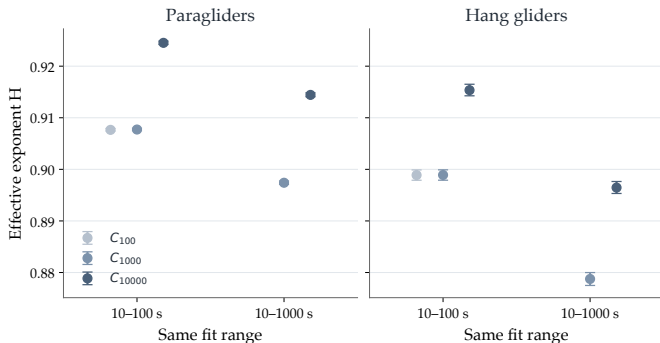
One cohort for 10–10,000 s

$$\tau_{\max} \leq 0.8T_s$$

Selected segments last
at least 12,500 s (3 h 28 min).

46,273 paragliders

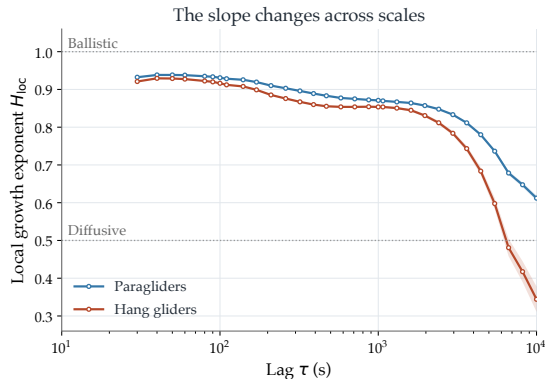
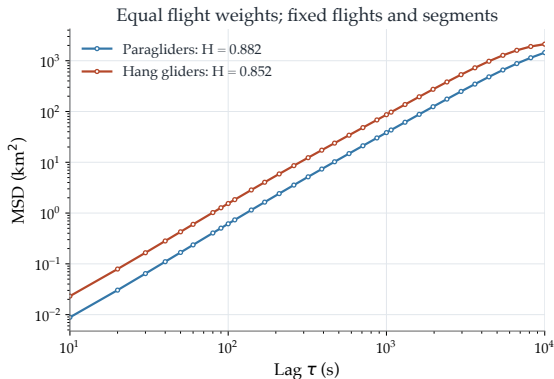
2,326 hang gliders



Selecting long segments raises H by about 2% on matched lag ranges

Fast growth persists, but one exponent is a summary

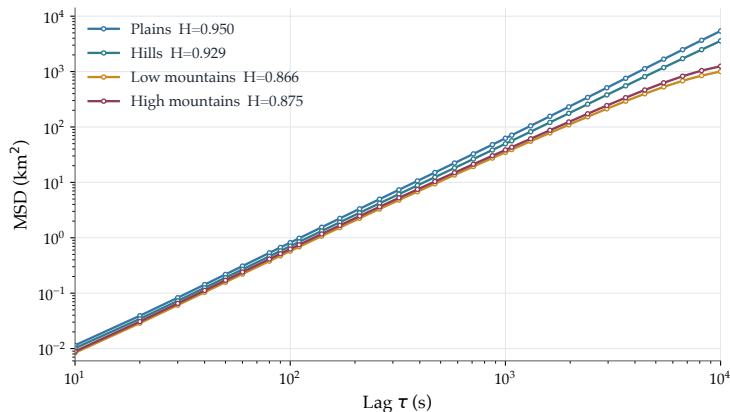
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Fixed population: $H_{para} = 0.882$ $H_{hang} = 0.852$ over 10–10,000 s

Launch-altitude groups have different MSDs

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Initial GNSS altitude

Plains: < 300 m

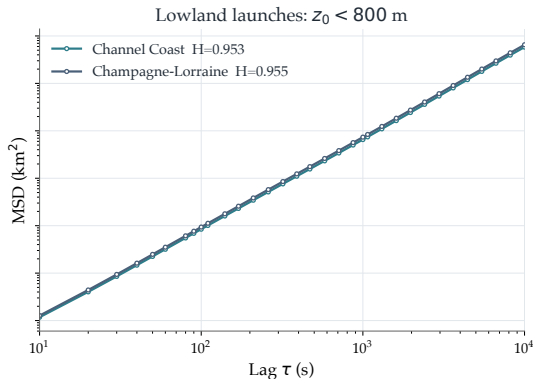
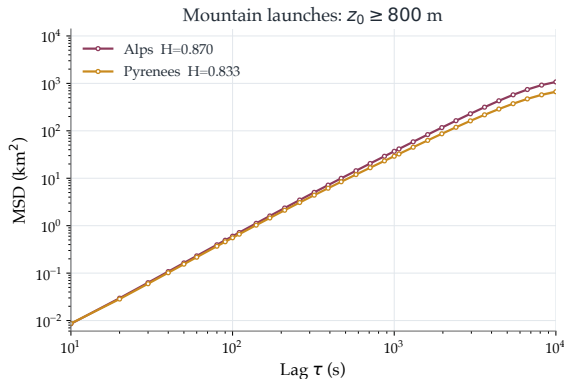
Hills: 300–800 m

Low mountains:
800–1,500 m

High mountains:
 $\geq 1,500$ m

Above sea level, at the first
cleaned fix.

Pooled lowland curves grow faster; circuit composition also changes with altitude



Alps–Pyrenees: $\Delta H = 0.037$

Lowland pair: $\Delta H = -0.002$, interval includes zero

Thermal patterns across height

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Viewer: intersect climb-labelled trajectories with horizontal planes.

Within each cell: same dates, segmentation and map bounds at both heights.

Mountains

Plains

$$z_{AGL} = z_1$$

$$z_{AGL} = z_2$$

$$z_{AGL} = z_1$$

$$z_{AGL} = z_2$$

Mountain cell
lower plane

Viewer screenshot

Mountain cell
higher plane

Viewer screenshot

Plains cell
lower plane

Viewer screenshot

Plains cell
higher plane

Viewer screenshot

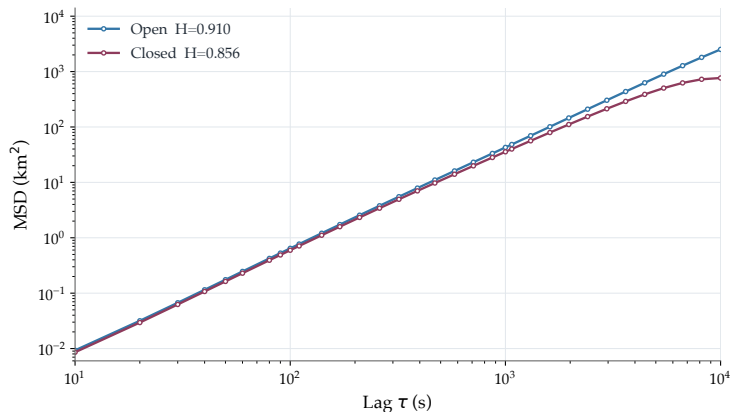
Working observation: mountain patterns shift less with height than lowland patterns.

Possible explanations: stronger uplift or weaker wind; neither is measured here.

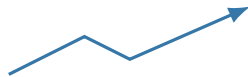
Dots are climb crossings, not thermal centres. z_{AGL} is relative to a fixed reference for each cell.

Open routes have larger displacement growth

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Open circuits



17,961 flights

Closed circuits

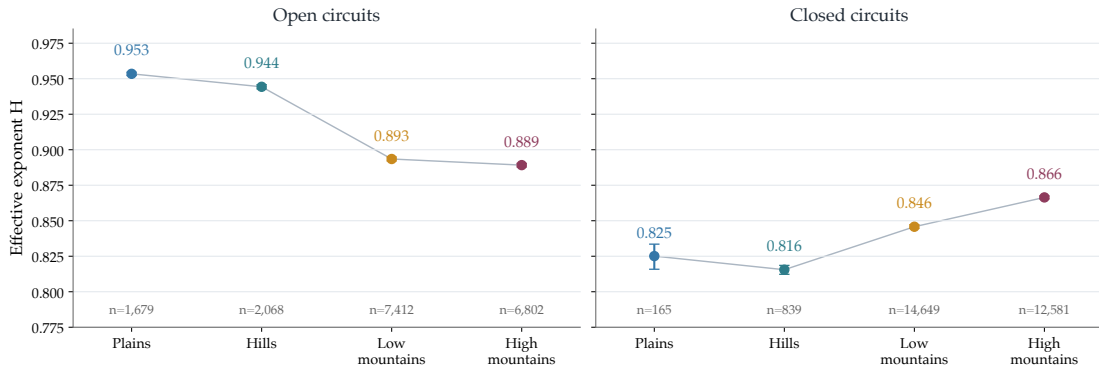


28,234 flights

$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] \quad (90\% \text{ paired interval})$$

The altitude ordering reverses for closed routes

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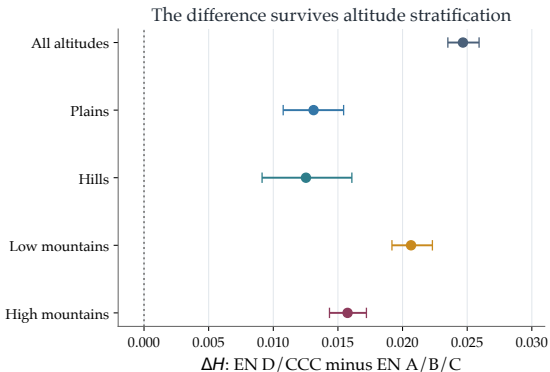
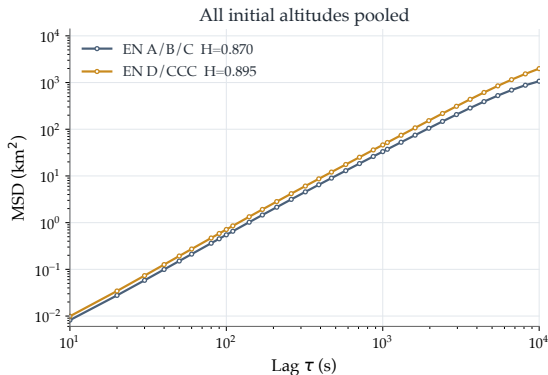


Open: lowland H is higher

Closed: mountain H is higher

The equipment contrast survives altitude stratification

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Higher H for EN D/CCC in every altitude class; largest gap in Low mountains

- 01 Rapid horizontal spreading in both disciplines, with an effective exponent that changes across scales.

- 02 The altitude association depends on the declared circuit: the lowland–mountain ordering reverses for closed routes.

- 03 Equipment-group differences remain within altitude classes; their physical and behavioural contributions are still mixed.

Next: explain global transport through flight behaviour and physical models